Valorization of rice husk biochar as a dual-confined slow-release fertilizer: Synergistic effects of pore-throat constriction and active-site engineering

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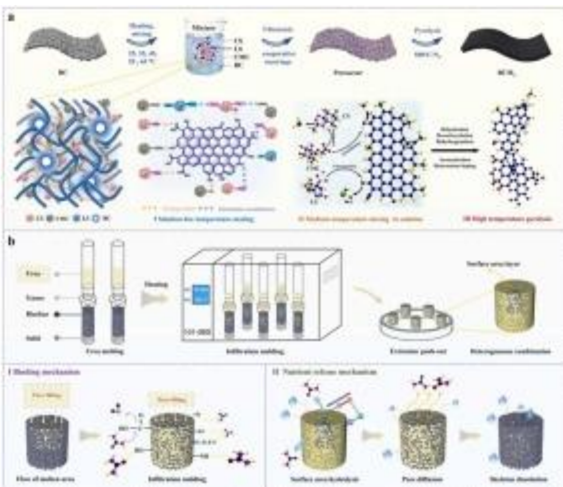
Highlights

- Ultrasonic modification achieved N/S co-doping and pore neck constriction.
- Urea adsorption boosted to 103.70 mg g^{-1} via a chemical-physical dual lock.
- Melt-infiltration reduced initial urea release rate by over 5.065-fold.
- Narrow pores and active sites effectively inhibited nutrient burst release.
- Large pore widths and defects were identified as key factors inducing leakage.

Abstract

Low nitrogen-use efficiency of urea fertilizers contributes to nutrient loss and environmental pollution. Here, rice husk biochar was engineered by a polymer-assisted ultrasonic cavitation strategy to realize N/S co-doping and pore-neck constriction, thereby enhancing nutrient adsorption and retention. The modification markedly optimized the biochar microstructure (increasing the specific surface area by nearly 8-fold and narrowing pore necks from 28.408 to 2.902 nm), while enriching adsorption-active sites, with pyridinic N reaching 51.10% and oxidized sulfur species increasing to 53.63%. Consequently, the urea adsorption capacity of the engineered biochar increased from 78.50 to 103.70 mgg^{-1}. Following capillary infiltration and solidification encapsulation with molten urea, the resulting fertilizer (BCM45UF) demonstrated superior controlled-release performance. Leaching tests revealed that BCM45UF effectively suppressed the initial burst release, reducing the initial release rate by over 5-fold compared to pure urea, and extended the effective release duration from 2 to 7 cycles. Kinetic and correlation analyses confirmed that this “burst inhibition, plateau maintenance, and long-tail extension” profile is synergistically governed by structure and chemistry: large pores induce early leakage, whereas narrow pore necks and dense heteroatom sites sustain long-term release. These findings provide a scalable, coating-free route to design high-efficiency biochar carriers that mitigate nitrogen loss.

Graphical Abstract



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Introduction

Owing to its high nitrogen content, low cost, and broad applicability, urea has long accounted for the majority of global nitrogen fertilizer consumption [1]. However, under conventional application practices, rapid urease-catalyzed hydrolysis leads to NH_3 volatilization, nitrification and denitrification generate reactive nitrogen emissions, and NO_3^- is leached with percolating water. Together, these processes result in persistently low nitrogen use efficiency in the field and give rise to environmental risks, including soil degradation, groundwater eutrophication, and greenhouse gas emissions [2]. Consequently, enhancing urea retention in soils, suppressing early burst release, and prolonging the period of steady nutrient supply through material innovation and process optimization, thereby improving nutrient availability to crops [3], is considered critical for achieving fertilizer input reduction while maintaining efficiency and for controlling agricultural non-point source pollution. To alleviate the problem of rapid nutrient loss caused by the fast dissolution of urea, slow- and controlled-release fertilizer technologies have been developed. Nevertheless, existing polymer- or sulfur-coated controlled-release fertilizers, while capable of moderating nutrient release to some extent, still suffer from several drawbacks, including high production costs, sensitivity to temperature and moisture fluctuations, susceptibility to burst release after heavy rainfall, and poor degradability of coating materials that can generate microplastic residues [4]. These limitations constrain their large-scale deployment in green and sustainable agricultural systems. Recently, significant advancements have been made in bio-based coating systems to mitigate these environmental concerns. For instance, dynamically crosslinked coatings, such as silicate-phenolic networks, have been elegantly engineered to encapsulate pure urea granules, achieving excellent controlled-release profiles while maintaining a high proportion of primary nutrients [5]. However, despite their high nutrient density and improved biodegradability, coating-based systems primarily serve a singular, transient function as nutrient carriers. In contrast, matrix-based CRFs present a paradigm shift towards dual-functionality. Although the absolute mass fraction of nitrogen in matrix CRFs is inherently lower than that of pure coated fertilizers, the supporting matrix provides irreplaceable industrial and agricultural value. Rather than acting merely as a temporary shell that eventually degrades, the matrix functions as a permanent soil amendment that continuously contributes to carbon sequestration, soil water retention, and the stimulation of microbial communities.

Against this background, biochar, characterized by high thermal stability, tunable pore structure and surface chemistry, and abundant feedstock availability, has been regarded as an environmentally friendly carrier for urea slow release. However, unmodified biochar typically suffers from several drawbacks, including a limited density of polar sites, wide and highly interconnected pore throats, and pore blockage and ash coverage [6]. As a result, when slow-release fertilizers first come into contact with water, urea is prone to capillary back-flow and outward seepage, leading to unstable nutrient release behavior and pronounced burst-release events. To further enhance the loading capacity and controlled-release performance of biochar as a nutrient carrier, various modification strategies have been proposed. For example, chemical activation with H_3PO_4, KOH [7], [8], and other activating agents can markedly increase the specific surface area and introduce oxygen-containing functional groups; N/S heteroatom doping [9] can be used to tailor surface polarity and electronic structure; and constructing organic interfacial layers on biochar surfaces using degradable polysaccharides or lignin-derived materials [10] helps strengthen hydrogen bonding and multipoint adsorption. Zhang et al. [11] demonstrated that ultrasound-induced cavitation can, under mild conditions, exfoliate carbon layers, remove ash and pore blockages, and open previously inaccessible micropores. Moreover, it can act synergistically with functional modifications involving urea, phosphoric acid, and related agents to enhance interfacial interactions [12]. These modification strategies have achieved notable progress in increasing specific surface area and enriching surface functional groups; however, their ability to finely tune the geometry of pores and pore throats remains limited.

To address the aforementioned challenges, we propose a long-chain polymer–assisted, ultrasound-induced cavitation strategy for pore engineering of rice husk biochar. Unlike conventional surface coating or simple chemical activation, this approach is designed to finely regulate the internal microenvironment of biochar by coupling physical cleaning with chemical reconstruction. Specifically, sodium lignosulfonate (LS), sodium carboxymethyl cellulose (CMC), and chitosan (CS) were selected as polymeric precursors because they provide complementary heteroatom sources and interaction motifs for in-pore anchoring and subsequent in situ carbonization. LS, a sulfonated lignin derivative, serves as an abundant S-containing aromatic precursor and improves wetting/dispersion, facilitating penetration into pore channels; CMC contributes carboxyl- and hydroxyl-rich chains that enhance polarity and can increase interfacial hydrogen bonding [10]; CS supplies amine-N and, under mild acidic conditions, becomes protonated and readily soluble, enabling strong electrostatic association with anionic LS/CMC.

Accordingly, a 2% (v/v) acetic-acid medium was used to solubilize and protonate CS, while simultaneously promoting the formation of an anionic–cationic polyelectrolyte complex/network with LS and CMC. Based on our preliminary formulation screening, LS, CMC, and CS were selected because their complementary functional groups and charge characteristics enable stable precursor complexation, effective pore infiltration, and subsequent in situ N/S incorporation during secondary pyrolysis [9]. The mass ratio of LS:CMC:CS was set at 1.0:0.6:0.4 according to these preliminary optimization results, balancing the S and aromatic-carbon contribution from LS, the viscosity/film-forming behavior and O-functional supply from CMC, and the N supply and electrostatic bridging provided by CS, thereby yielding a precursor solution that remains readily infiltratable yet forms a sufficiently robust in-pore polymeric network for pore-throat reshaping after pyrolysis [13].

Considering that the migration of biochar powders in soils may pose risks to aquatic ecosystems [11], we further adopt a molten-urea capillary-infiltration and solidification-encapsulation strategy to process the modified biochar into shaped slow-release fertilizers. This design integrates geometric confinement with chemical binding, thereby enabling controlled nutrient release without relying on thick external coatings, in line with the preparation concept reported by Xiang et al. [13].

Building on the above objectives, rice husk biochar was employed as the base material and loaded with polymeric precursor solutions at different mixing temperatures. Polymer-loaded modified biochars (BCMx) were obtained by combining ultrasound treatment, vacuum-assisted impregnation, and secondary pyrolysis. We systematically evaluated how these modifications regulate the density of polar adsorption sites and the pore-throat architecture, and how the resulting structural changes affect urea adsorption kinetics, isotherm behavior, and their temperature dependence. In parallel, the modified biochars were processed into shaped fertilizers using a molten-urea infiltration–solidification process, and sand-column leaching experiments coupled with kinetic model fitting were carried out. By linking the release profiles with the physicochemical properties of the biochars, we elucidated the structural origins of burst-release suppression, plateau maintenance, and extended long-tail release. Overall, this work establishes a materials-design concept and processing pathway for the development of scalable, environmentally benign, biochar-based controlled-release nitrogen fertilizers that do not rely on poorly degradable coatings.

Materials and chemical reagents

Rice husks were obtained from Hefei (Anhui, China). The fundamental physicochemical properties of the raw material were determined as follows: C 35.24%, H 5.03%, O 39.56%, N 0.84% (atomic ratios: H/C 1.71, O/C 0.84), with an ash content of 19.33% and a pH of 6.47. Chitosan (CS, biochemical grade, degree of deacetylation 80–95%), sodium lignosulfonate (LS, analytical grade), sodium carboxymethyl cellulose (CMC, analytical grade), acetic acid (CH_3COOH, 99.5%, AR), urea (CH_4N_2O, 99%), and

Characterization of the modified biochars

The influence of mixing temperature on the microscopic morphology of the modified biochar is visually captured in the SEM images (Fig. 3A). Increasing the temperature from 25 to 65℃ drives a distinct morphological transition of the pore filler: from discrete particles (BCM25/35) to a semi-continuous rough film (BCM45), and finally to a wrinkled, porous thick film at 55–65℃ (BCM55/65). This evolution is governed by solution rheology; higher temperatures lower viscosity while enhancing wetting

Conclusion

This study successfully developed a modified rice husk biochar via a polymer-assisted ultrasonic strategy, achieving simultaneous N/S co-doping and pore structure optimization.

The modification significantly increased the specific surface area from 5.423 to 40.461 m^2g^{-1} and narrowed the pore necks from 28.408 to 2.902 nm. Chemically, the surface was enriched with high-density polar sites, with Pyridinic-N and Oxidized-S contents reaching 51.10% and 53.63%, respectively. Consequently, the urea

CRediT authorship contribution statement

Mingfeng Wang: Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Donghai Wang:** Writing – review & editing, Supervision, Methodology. **Dongni Qiu:** Data curation. **Jie Li:** Data curation. **Yuyang Cong:** Writing – original draft, Data curation, Conceptualization. **Zhiwen Chen:** Writing – review & editing, Supervision, Methodology. **Ke Zhang:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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